



OPEN Quantitative assessment of human-exoskeleton integration through a neurophysiological marker of embodiment

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Assistive exoskeletons are increasingly used in rehabilitation settings, yet their adoption in real-world scenarios remains limited. This is partly due to the insufficient understanding of human-device interaction mechanisms. The process through which the central nervous system integrates external devices is known as “tool embodiment” and is primarily assessed through subjective questionnaires. This protocol aims to identify a quantitative marker of embodiment by leveraging the phenomenon of sensory attenuation, i.e., the modulation of sensory stimulation intensity perception as a function of action control. Twenty-one participants either self-administered electrical stimulation to the median nerve by stepping on a pedal or remained still during stimulation by the experimenter, in two conditions: with and without wearing a lower-limb exoskeleton. Somatosensory evoked potentials (SSEPs) were recorded using a 64-channel electroencephalography. Results showed reduced N60 amplitude during self-stimulation, consistent with sensory attenuation, in both conditions. However, wearing the exoskeleton led to higher N60 amplitudes, suggesting reduced motor prediction accuracy. External stimulation elicited a larger N100 component, likely reflecting increased attentional engagement. Together, N60 and N100 serve as promising neurophysiological indices of embodiment and cognitive load. These objective markers could be integrated into the design and evaluation of robotic devices, improving usability and fostering better device-user integration.

Keywords Electroencephalography, Embodiment, Human-robot interaction, Somatosensory evoked potentials

Exoskeletons are increasingly being adopted in rehabilitation settings as they allow intensive, task-specific, and multisensory stimulation¹. However, several barriers remain to their effective uptake in real-world scenarios. Current exoskeletons are often bulky and difficult to wear and control, leading users to perceive them as external devices rather than extensions of their bodies². These limitations hinder their acceptance, integration into daily life, and overall efficacy of rehabilitation³. Previous research by our group showed neurophysiological activations (both cortical – EEG – and muscular – EMG) that were not entirely consistent with a physiological gait pattern while wearing a lower limb exoskeleton^{4–6}. This observation raised the question of human-robot integration during gait training with exoskeletons. The theoretical framework of embodiment provides a compelling lens through which to explore these complex integrative processes. Therefore, investigating factors that could improve the efficiency of human-exoskeleton integration is crucial. The integration of external objects as part of the human body has been referred to as tool embodiment⁷. This process is associated with the plastic capacity of the CNS to temporarily incorporate objects within the body schema, which is the sensorimotor representation of the body and emerges from multisensory integration processes⁸. Once a tool is embodied, three dimensions of body perception are consequently affected: the sense of ownership (i.e., the feeling that the tool is part of one's own body), the sense of agency (i.e., the feeling of control over the tool's movements), and the sense of self-location (i.e., the perceived spatial position of the body or body parts)^{9,10}.

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The brain mechanisms underlying tool embodiment have previously been explained by the feedforward theory of movement control¹¹. According to this model, every time a motor program is sent from the primary motor cortex (M1) to the motor neurons in the spinal cord, a copy of the descending signal is generated in parallel (i.e., efference copy), allowing the nervous system to predict the sensory consequences of the programmed movement. The efference copy and the afferent signals generated from the movement execution are compared to track errors and enable motor adjustments. The more expected movement and actual movement match (i.e., spatiotemporal congruency), the more the subject experiences a sense of agency over that movement, a sense of self-location with the body part or tool used, and a sense of ownership over the action. In contrast, a consistent mismatch will lead to a decreased tool embodiment (see Fig. 1). Through the efference copy, the CNS predicts the sensorimotor consequences of our actions, leading to the phenomenon of sensory attenuation. Sensory attenuation is built on a reduced neurophysiological response to sensations generated by our own actions, which are predictable, compared to identical sensations produced by external sources^{12,13}. This mechanism aids in detecting biologically significant and less predictable external stimuli by filtering out predictable information generated by our own motor activity¹³. Sensory attenuation has been measured by the electroencephalographic (EEG) signal as a reduction in the amplitude of self-produced sensory evoked potentials (SEPs) with respect to those externally generated. These effects have primarily been studied in the auditory modality: self-generated sounds generate an attenuated potential (i.e., N1-P1 complex) compared to externally generated ones^{14–16}. However, other sensory modalities have been explored. For instance, previous studies have investigated the somatosensory modality through paradigms involving non-painful electrical stimulations delivered to the median nerve^{13,17}. These experiments show significant attenuation of early and mid-latency somatosensory evoked potentials (SSEP) components (i.e., N20 and N60) over the contralateral sensorimotor cortex during self-induced stimulation, compared to external stimulation^{13,18}.

In this study, we leverage sensory attenuation to assess the integration between the user and a lower limb exoskeleton. Until now, exoskeleton embodiment has been largely overlooked, and most assessments are based on self-reported questionnaires³. A neurophysiological marker to measure the integration between humans and exoskeletons is currently lacking, yet it might be highly valuable for evaluating exoskeleton designs and enhancing their ability to assist human movement. To address this gap, we used EEG measurements of SSEPs to quantify sensory attenuation induced by self-generated versus externally generated electrical stimulation while participants wore a lower limb exoskeleton (see Fig. 2). Additionally, participants provided self-reported ratings of perceived stimulation intensity at the end of each trial as a subjective dimension of embodiment. In particular, we tested the following hypothesis: the sensory attenuation induced by self-generated actions compared to externally generated actions is modulated by the embodiment between the user and the exoskeleton; e.g. if the exoskeleton is embodied, self-induced stimulation should not differ from self-induced stimulation without exoskeleton.

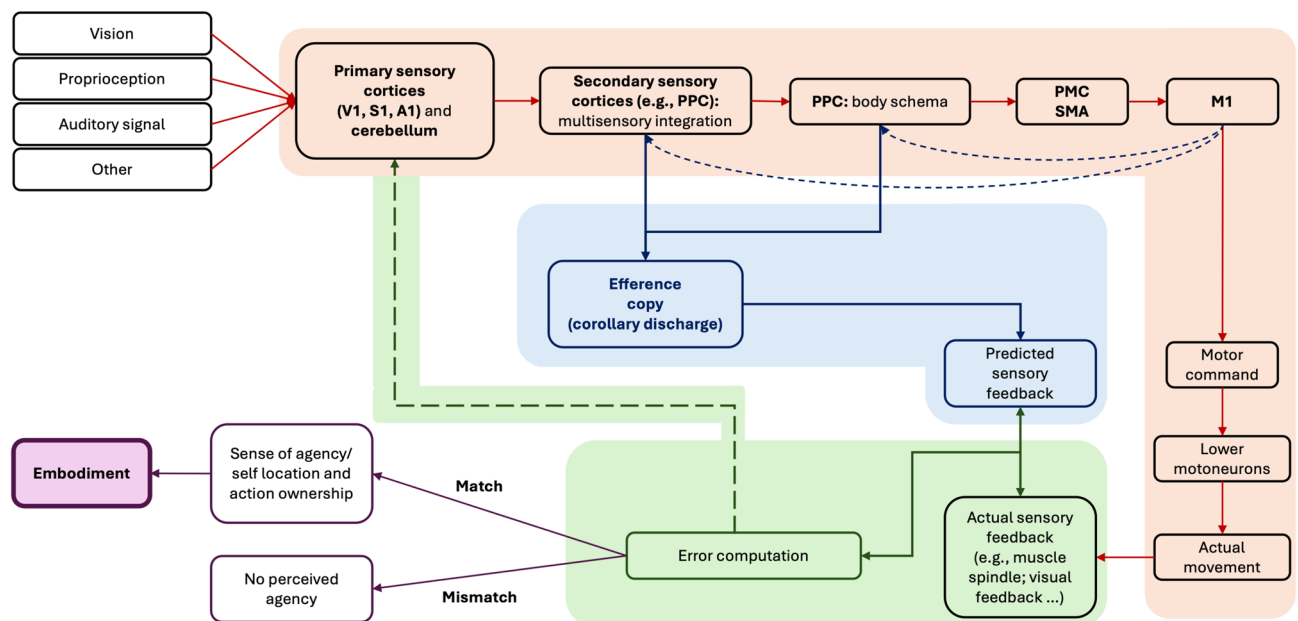


Fig. 1. The schema illustrates the proposed mechanism of embodiment. The red-shaded areas represent the voluntary movement execution pathway; the blue-shaded areas indicate feedforward control; the green-shaded areas correspond to feedback control. Dotted lines represent feedback connections. Abbreviations: V1= primary visual cortex; S1= primary sensory cortex; A1: primary auditory cortex; PPC: posterior parietal cortex; PMC: premotor cortex; SMA: supplementary motor area; M1: primary motor cortex.

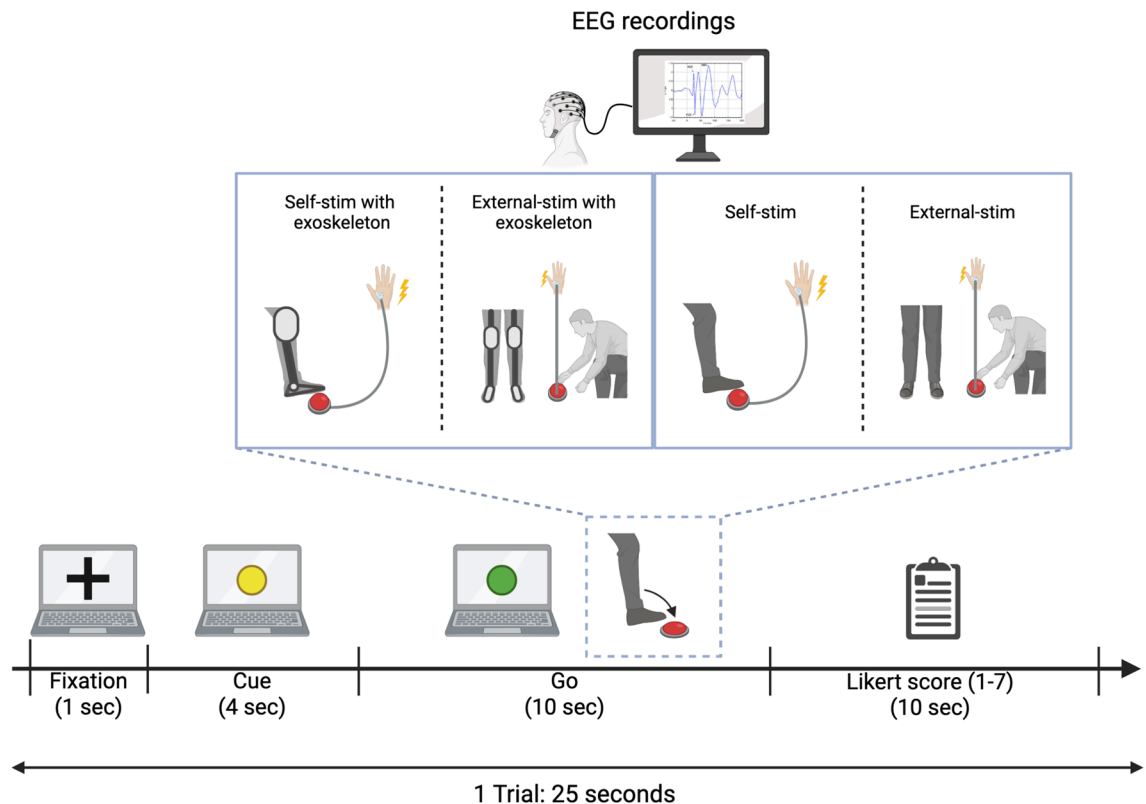


Fig. 2. The cartoon represents the sequence of events in each trial. The cue dot could be yellow, blue, or pink, representing the self, external, or mixed stimulation conditions, respectively. The same sequence of events was presented in both the exoskeleton and non-exoskeleton blocks.

Results

Perceived stimulation intensity ratings

Stimuli intensity ratings were normally distributed (see supplementary materials, table S1). Repeated measure ANOVA showed a significant main effect of the stimulation type (i.e., self vs. external vs. mixed) when donning the exoskeleton, ($F=75.27$, $p<0.001$). Post hoc pairwise t-test showed that participants rated externally generated stimuli as significantly more intense than self-generated ones (mean self-stimulation score: 3.01 ± 0.81 , mean external stimulation score: 4.32 ± 0.78 , $p<0.001$). The mixed-stimulation condition yielded intermediate ratings (mean mixed stimulation score: 3.17 ± 0.77) significantly different from the self-stimulation ($p=0.022$) and the external stimulation conditions ($p<0.001$). In conditions without exoskeleton, the stimulus intensity ratings did not differ significantly between conditions (mean self-stimulation score: 3.75 ± 0.69 , mean external stimulation score: 3.99 ± 0.81 , $p=0.158$). See Fig. 3. This pattern suggests that wearing the exoskeleton modulates the subjective perception of stimulation intensity, increasing the perceived contrast between self-generated and externally generated stimuli.

Sensory attenuation: N60 component

The amplitudes of the N60 component were not normally distributed (see supplementary materials table S2). To assess the presence of sensory attenuation in the control condition, we first compared the self and externally generated stimulation without the exoskeleton (i.e., Block B). As shown in Fig. 4, the Wilcoxon signed rank test for the paired samples confirmed the presence of sensory attenuation: N60 amplitudes were significantly reduced during self-stimulation (median amplitude of N60 for self-stimulation: $-0.90 \mu V$ [-3.22 , 0.58]) compared to external stimulation ($-2.07 \mu V$ [-7.64 , 0.12]; $V=217$, $p<0.001$).

We then examined whether this effect persisted while wearing the exoskeleton (Block A). The Friedman's rank sum test showed a significant effect (Friedman chi-squared = 19.14 , $p<0.001$). Figure 5 illustrates the significant post-hoc comparisons between self and external stimulation (median N60 amplitude for self-stimulation: $-1.5 \mu V$ [-4.66 , 0.40]; median N60 amplitude for external stimulation: $-2.49 \mu V$ [-7.79 , 0.06]; $p=0.001$), and between mixed and external stimulation (median N60 amplitude for mixed stimulation: $-1.25 \mu V$ [-5.3 , 0.23]; $p<0.001$).

To assess whether the presence of the exoskeleton modulated the sensory attenuation effect, we compared the N60 amplitudes between the exoskeleton conditions within each stimulation type (e.g., self-stimulation with the exoskeleton vs. self-stimulation without the exoskeleton). The Wilcoxon test revealed a significant difference in N60 amplitude during self-stimulation, with smaller N60 amplitude in the no-exoskeleton condition. Specifically, the median N60 amplitude for self-stimulation with the exoskeleton was $-1.50 \mu V$ [-4.66 , -0.40], whereas the

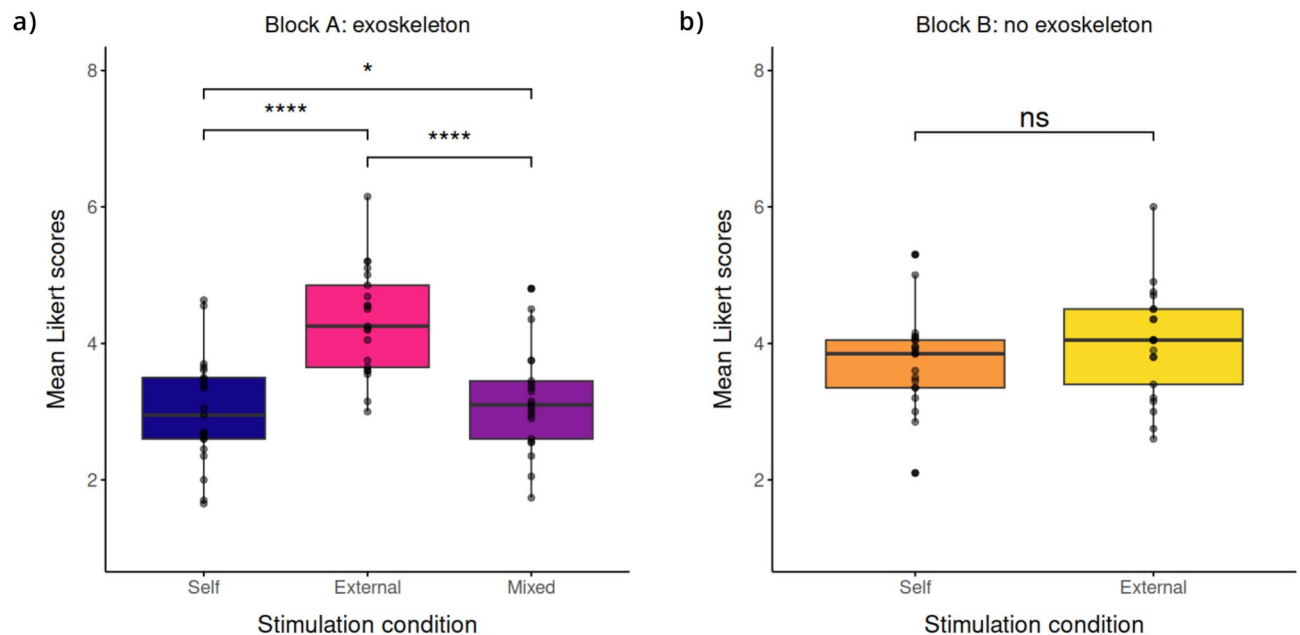


Fig. 3. (a) Mean ratings of perceived current intensity in the three stimulation conditions while wearing the exoskeleton: self-stimulation (i.e., blue boxplot), external stimulation (i.e., pink boxplot), and mixed stimulation (i.e., purple boxplot). (b) Mean scores for the block without the exoskeleton: self-stimulation (i.e., orange boxplot) and external stimulation (i.e., yellow boxplot). (ns): $p > 0.05$; (*): $p < 0.05$; (****): $p < 0.0001$.

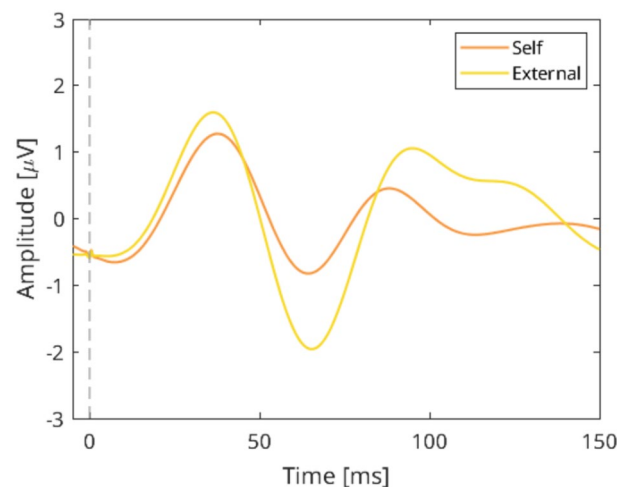


Fig. 4. The plot shows the N60 grand average amplitudes at CP4 for the self (i.e., orange line) and external (i.e., yellow line) stimulation conditions while not wearing the exoskeleton.

median amplitude for self-stimulation without the exoskeleton was $-0.90 \mu V$ [$-3.22, 0.58$] ($V = 192, p = 0.006$). A similar trend was observed comparing the N60 amplitudes during external stimulation conditions. The median N60 amplitude for external stimulation with the exoskeleton was $-2.49 \mu V$ [$-7.79, 0.06$], whereas the median amplitude for external stimulation without the exoskeleton was $-2.07 \mu V$ [$-7.64, 0.12$] ($V = 180.5, p = 0.025$).

Finally, we assessed interaction effects by comparing: 1) self-stimulation without the exoskeleton vs. external stimulation with the exoskeleton, and 2) self-stimulation with the exoskeleton vs. external stimulation without the exoskeleton. As expected, both comparisons were significantly different (respectively, $V = 227, p < 0.001$; $V = 179, p = 0.026$). See Fig. 6.

Sensory attenuation: N100 component

The amplitudes of the N100 component were normally distributed (see the supplementary materials, table S3). A clear negative peak around 100 ms after external stimulation was observed in Cz, regardless of exoskeleton. In

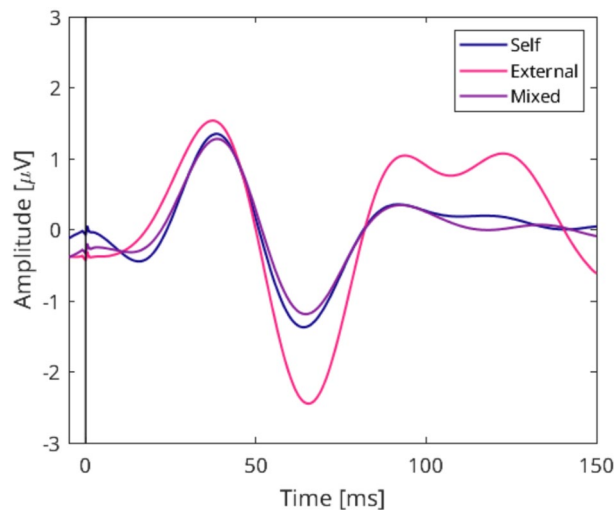


Fig. 5. The plot shows the N60 grand average amplitudes at CP4 for the self (i.e., blue line), external (i.e., pink line) and mixed stimulation (i.e., purple line) conditions while wearing the exoskeleton.

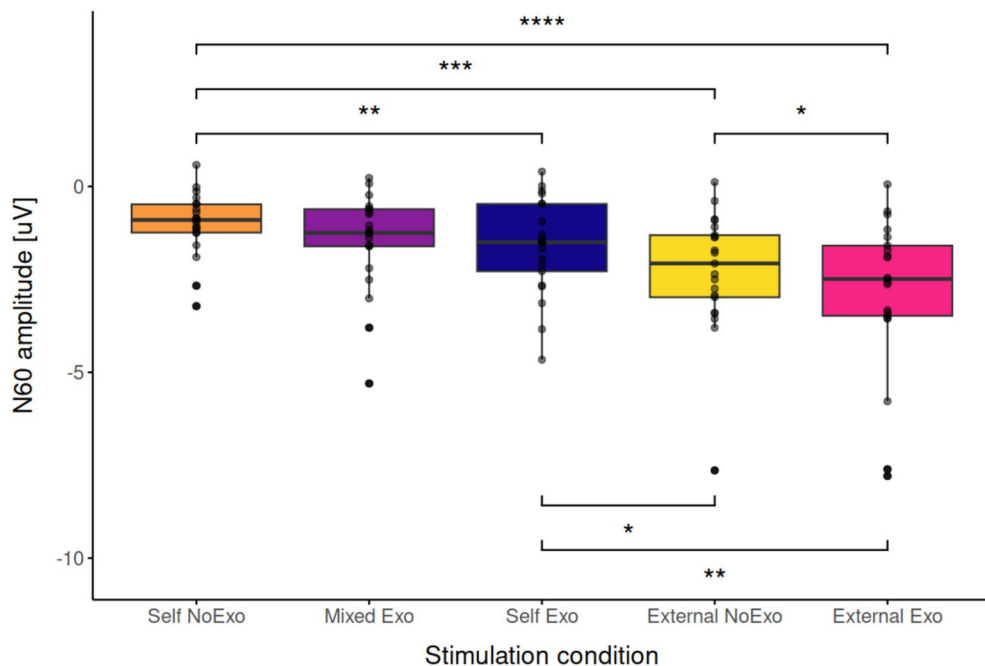


Fig. 6. The boxplots are arranged from the lowest N60 amplitude (self-stimulation without exoskeleton) to the highest amplitude (external stimulation with exoskeleton). The figure highlights how the N60 amplitude is modulated by the presence of the exoskeleton within the same stimulation condition. (ns): $p > 0.05$; (*): $p < 0.05$; (**): $p < 0.01$; (***): $p < 0.001$; (****): $p < 0.0001$.

Block A (with the exoskeleton), the ANOVA was significant ($F = 56.04$, $p < 0.0001$), and post hoc tests revealed significant differences between self-stimulation and external stimulation (mean N100 for the self-stimulation: $1.04 \pm 0.93 \mu V$; mean N100 for the external-stimulation: $-5.17 \pm 2.89 \mu V$, $p < 0.0001$), as well as between external and mixed stimulation conditions (mean N100 for the external-stimulation: $-5.17 \pm 2.89 \mu V$; mean N100 for the mixed-stimulation: $-1.00 \pm 0.94 \mu V$, $p < 0.0001$).

A similar pattern was observed in Block B (without the exoskeleton), where the N100 amplitude for external stimulation was significantly higher than for self-stimulation ($t = 6.79$, $p < 0.0001$; see Fig. 7). However, when comparing the same stimulation types across exoskeleton conditions (e.g., self with vs. without exoskeleton), no significant differences were found. These results confirm that the N100 component differentiates self- from externally generated stimulation, and that this effect remains stable across exoskeleton conditions.

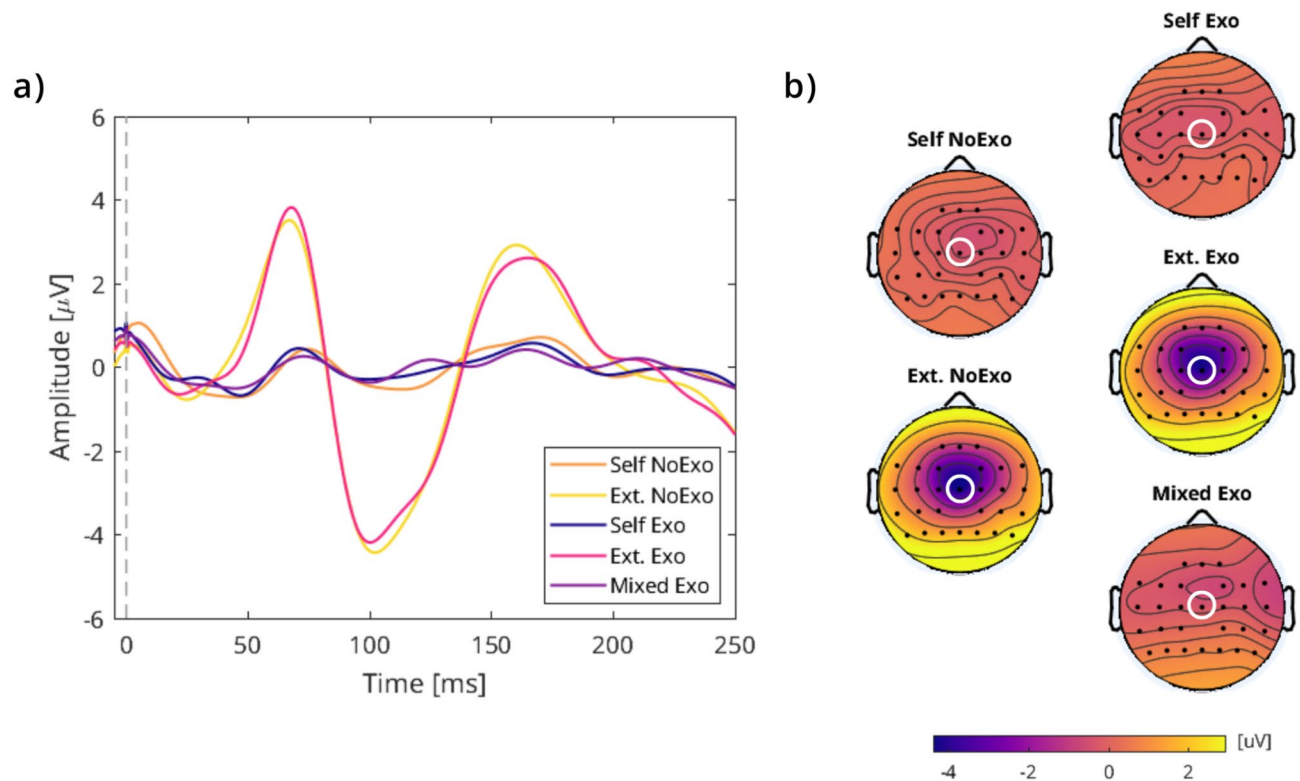


Fig. 7. (a) The figure shows the N100 grand average amplitude at Cz across all experimental conditions, clearly highlighting a negative peak in the external stimulation conditions. (b) Topoplots of the N100 across all conditions. Blue color indicates more negative peaks.

	N60	N100	Likert
N60	1.00	0.58	-0.26
N100	0.58	1.00	-0.60
Likert	-0.26	-0.60	1.00

Fig. 8. The matrix displays the r values for multiple pairwise correlations between N60, N100 and the Likert scores.

Correlation between neurophysiological and subjective measures

Repeated measure correlation analysis was conducted to examine the relationships between the N100, N60, and the Likert scores (see Fig. 8). A significant positive correlation was found between the N60 and N100 amplitudes ($r = 0.58$, $p < 0.0001$). In contrast, negative correlations were observed between the N100 and Likert scores ($r = -0.60$, $p < 0.0001$), and to a lesser extent between the N60 and Likert scores ($r = -0.26$, $p = 0.017$).

Discussion

Our aim was to identify a neurophysiological quantitative marker of human-robot embodiment. The initial hypothesis was confirmed: the amplitude of the middle latency SSEP component (i.e., N60), triggered by median

nerve stimulation, was modulated by the presence of the exoskeleton. More specifically, N60 amplitudes over the somatosensory area were higher in the exoskeleton conditions compared to the corresponding non-exoskeleton conditions. Previous research showed that self-stimulation leads to sensory attenuation, i.e., a reduced amplitude of the corresponding event related potentials (ERPs)¹⁹. This finding was replicated in the present study, as evidenced by the sensory attenuation observed in the experimental conditions that did not involve the exoskeleton. Recent research also showed that sensory attenuation can occur when the stimulation is mediated by an embodied object (e.g., a fake hand), but not when the object is not embodied¹⁹.

In our study, we did not expect to observe embodiment between the exoskeleton and the user, given the characteristics of the exoskeleton (e.g., bulkiness, rigidity) and the lack of prior familiarization with it²⁰. However, sensory attenuation, albeit with a lesser degree than in the non-exoskeleton condition, was still observed when wearing the exoskeleton. We argue that this likely reflects the differences in the experimental conditions rather than true embodiment. Specifically, self-stimulation required participants to initiate a voluntary motor action, which, in accordance with motor control theories, generates predictions about the expected sensory consequences of that action. Although these predictions may be less precise in the exoskeleton condition, they are nonetheless sufficient to produce sensory attenuation²¹.

Importantly, when comparing across the same stimulation type, the N60 amplitudes were less attenuated when the participant wore the exoskeleton. This finding could be interpreted as a reduction in the participant's ability to plan precise movements while wearing the exoskeleton and self-stimulating, leading to a less pronounced sensory attenuation. This may occur because the user has not developed an accurate internal model of the exoskeleton's dynamics, making movements less precise and their outcomes more difficult to predict^{20,21}. On the other hand, the highest N60 peak observed in the external stimulation condition while wearing the exoskeleton could indicate a general alteration of body control perception, even when participants were not required to move. This suggests that wearing an exoskeleton could lead to a sense of disembodiment, even without movement. In fact, simply wearing an exoskeleton can alter proprioceptive feedback. This is in line with previous research showing that wearing exoskeletons or prosthetics can influence proprioception and body schema, with the experience of either embodiment or disembodiment depending on how the wearer interacts with the device²⁰. In this context, it is crucial to emphasize that factors such as the weight of the exoskeleton are not confounding variables, but rather a central component of the sensorimotor experience under investigation. Prior studies have shown that changes in sensory feedback resulting from physical properties like device weight can alter sensorimotor integration, thereby influencing the sense of embodiment^{20,22}. Therefore, the increased N60 amplitude may partly reflect the nervous system's response to disrupted proprioceptive predictions induced by the exoskeleton's physical characteristics. Regarding the mixed-stimulation condition, no differences were observed compared to the self-stimulation condition. The purpose of introducing this intermediate condition was to experimentally modulate the user's perceived control over the action by passively moving the user's leg through external activation of the exoskeleton. However, due to the characteristics of the actuators, our exoskeleton was unable to move the subjects passively without their active involvement. The ALICE exoskeleton is an assistive device, meaning its actuators do not fully support the movement of a healthy adult leg. As a result, the user's active participation is necessary to complete the movement, making this condition effectively comparable to the self-stimulation condition. EEG data align with participants' self-reported ratings of perceived stimulation intensity.

At the behavioral level, participants rated the perceived intensity as highest in the external stimulation condition, intermediate in the mixed stimulation condition, and lowest in the self-stimulation condition, despite the stimulation intensity being the same across all conditions. This effect, widely documented in the literature, seems primarily mediated by attentional processes^{23,24}. Notably, modulation of subjective stimulus intensity was significant only in the exoskeleton conditions. This effect may reflect a state of disembodiment: the greater mismatch between predicted and actual sensory input in these conditions likely increases demands on attentional resources. Supporting this, a qualitative study found that simply wearing a heavy exoskeleton makes users constantly aware of the device, heightening their overall sensory awareness²⁵. By contrast, in non-exoskeleton conditions, more accurate sensorimotor predictions reduce the need for such attentional shifts, leading to diminished modulation of perceived stimulus intensity. In our study, this is further supported by the strong negative correlation between self-reported intensity scores and the amplitude of the central N100. Research on attention orienting in the somatosensory modality has shown that the N100 is a marker of attention orientation to external somatosensory stimuli, which mediates the perception of stimulus intensity²⁴. In this study, the N100 was significantly more negative in the external stimulation conditions compared to self-stimulation, regardless of the presence of the exoskeleton. These results likely reflect the natural tendency to focus on external stimuli, which are perceived as more evolutionarily significant than self-generated stimuli and, as a result, are subjectively experienced as more intense. Lastly, a positive correlation was found between the N60 and N100. This is not surprising, as some authors have previously suggested that sensory attenuation is partly modulated by attentional processes, although attention alone cannot account for the entire phenomenon²³. Consequently, we argue that the marginal negative correlation observed between the N60 and the Likert scores is likely a spurious correlation, mediated by the N100.

In conclusion, the amplitude of the middle-latency N60 SSEP component is influenced by how predictable the results of an action are. Based on this knowledge, we proved that the N60 amplitude can act as an objective marker of human-robot embodiment. Specifically, the more similar the N60 amplitude following a self-generated action (without the exoskeleton) to the N60 amplitude generated with the exoskeleton, the greater the integration of the exoskeleton into the motor and sensory systems of the user.

Future research should further explore the potential of the N60 component as a quantitative neurophysiological marker of human-robot embodiment. Longitudinal studies could examine how repeated training or increased familiarity with the exoskeleton modulates sensory attenuation and embodiment over time. Additionally,

extending this approach to different types of exoskeletons or control strategies could clarify how specific mechanical and control parameters can influence the integration of the device into the user's body schema. Finally, translating these findings into clinical and ergonomic contexts should support the development of more intuitive, user-centered exoskeletons that promote embodiment and enhance motor rehabilitation outcomes.

Methods

This was an observational study conducted following the Declaration of Helsinki guidelines and approved by the Ethics Committee for Psychological Research of the Department of General Psychology of the University of Padova (code number: 163-b).

Participants

Twenty-one healthy volunteers (mean age: 24.62 ± 2.2 ; 10 females) of the Padova University took part in the study and provided written informed consent. The inclusion criteria were as follows: age over 18; no prior experience with exoskeletons; no diagnosed clinical motor and/or cognitive disorders that could affect task comprehension and/or performance; compliance with the anthropometric requirements of the exoskeleton (height range: 155–180 cm, weight: <90 kg); right-handed. Power analysis was performed using G * Power to determine the required sample size for the study, with an alpha level of 0.05, an effect size of 0.25, and a power of 0.80. The analysis indicated that the required sample size was 21 participants.

Material

The following instruments were used to conduct the experiments:

- The Digitimer DS7A electrical stimulator, employed to deliver non-painful electrical stimulation during each trial via bipolar Ag/AgCl surface electrodes placed on the participants' left median nerve. The intensity of stimulation was established, based on the previous literature¹⁹, at a 2.5-fold threshold for subjective stimulus detection, plus an additional 4 mA, with a voltage of 300 V and a pulse duration of 100 μ s. The subjective threshold was determined by gradually increasing the intensity of the stimulation until it was clearly perceived at least in half of the trials (i.e., 5 out of 10 trials). Once determined, the stimulation intensity was kept constant in all experimental conditions, unnoticed by the participants. The stimulation could be activated either by pushing a button on the Digitimer system or by pressing a pedal. When a stimulation was delivered, a TTL output trigger pulse (1 ms duration) was generated and sent to the EEG amplifier to synchronize the signals.
- EEG data was collected from a 64-channel electrodes cap and amplifier (eegoTM sports, ANT Neuro, Netherlands). The EEG electrodes were positioned according to the International 10–20 system²⁶. The reference electrode was located at the CPz electrode position, whereas the AFz electrode served as the ground electrode. The ROS-Neuro middleware²⁷ was used to acquire, record, and synchronize the EEG signals with a sample rate of 2048 Hz. Synchronization with the electrical stimulator was achieved with the recording of the TTL signal through the dedicated input trigger port of the amplifier.
- The ALICE lower-limb assistive exoskeleton (INDI Engineer and Design, France) was used as a wearable robotic device during the experiments. The ALICE exoskeleton is an open-source platform (<https://hackaday.io/project/176681-alice-open-source-exoskeleton-2021-update>) that we constructed and used in a previous study on human-robot integration². The exoskeleton features four active joints (i.e., two hips and two knees) powered by brushed DC motors and a worm gearbox. Each actuation unit provides a maximum torque of 26 Nm and a nominal torque of around 10 Nm. It operates at a rated voltage of 12 V and consumes 120 W of nominal power, with peak power consumption reaching up to 260 W. The low-level control unit (i.e., Arduino Mega 2650 R3, Arduino, Italy), high-level control unit (i.e., Beelink SEi8, Intel® Core i5, 8 GB RAM, Beelink, Shenzhen, China), and the power supply are housed off-board and connected to the exoskeleton via a 15-meter cable bundle. Each joint is equipped with a linear encoder to measure the joint position in real time. The lengths of the links and the dimensions of the pelvis frame were manually adjusted by a skilled operator to fit the body characteristics of each participant. The low-level controller of the exoskeleton consists of a PI position control with velocity constraints ($K_p = 2.0$, $K_i = 3.0$ for hip joints; $K_p = 3.0$, $K_i = 5.0$ for knee joints) that moves the exoskeleton's joints along a pre-defined trajectory when the step motion is activated. The exoskeleton's gait profile could be triggered in the high-level controller in two ways, either manually by an operator from the dedicated graphical interface, or voluntarily by the participants by initiating a step with the right leg. Once activated, the exoskeleton supplemented the user's leg strength by providing additional power to assist in completing the gait trajectory.

Data collection

The protocol used in this study was a modified version of a previous one used to assess sensory attenuation¹⁹. The experimental protocol consisted of two blocks, presented to each participant in a randomized order. In Block A, participants performed the task while wearing the lower limb exoskeleton; in Block B, the same task was performed without the exoskeleton. Within each block, participants were asked to either stimulate their median nerve by stepping (i.e., single step) on a switch located on the floor and connected to the DS7A (i.e., self-stimulation condition) or remain still and wait for the experimenter to deliver the stimulation (i.e., external stimulation condition). In Block A, we included a third stimulation condition, called mixed stimulation. In this condition, the experimenter partially assisted the participant by activating the exoskeleton to guide the leg toward the switch, while the participant still had to actively complete the step to trigger the stimulation. This design provided an intermediate level of action control, lying between fully self-generated stimulation (participant initiates and executes the action independently) and fully externally generated stimulation

		Between subjects' randomization	
		Exoskeleton (A)	No-Exoskeleton (B)
Within subjects' randomization	Self-stimulation (1)	1 A (20 trials)	1B (20 trials)
	External-stimulation (2)	2 A (20 trials)	2B (20 trials)
	Mixed-stimulation (3)	3 A (20 trials)	/
		60 trials TOT	40 trials TOT

Table 1. Experimental conditions and randomization.

(experimenter triggers the stimulation without participant involvement). This allowed us to examine how partial control influences both neurophysiological and subjective measures of embodiment. See Table 1 for a summary of the experimental conditions and the related number of trials. Each trial was structured as follows: Participants were asked to look at a screen in front of them. A fixation cross appeared for 1 second, followed by a 4-second cue. The cue was a colored dot that indicated the upcoming stimulation condition: a yellow dot for self-stimulation, a blue dot for external stimulation, and a pink dot for mixed stimulation. Then a go-cue, represented by a green dot, appeared on the screen for 10 seconds. During this interval, the participant was required to step on the button or the experimenter could activate the stimulation, depending on the cue. At the end of each trial, participants verbally rated the perceived stimulation intensity on a Likert scale from 1 to 7, where 1 represented very low intensity and 7 represented very high intensity. These subjective ratings served as behavioral measures of sensory attenuation, which we adopted as an indirect index of neurophysiological embodiment^{28,29}. See Fig. 2 for a schematic representation of the experimental procedure.

Data analysis

EEG data were processed using MATLAB (Version R2024b) and the EEGLAB toolbox (Version v2024.2). The preprocessing pipeline was designed to reduce artifacts and enhance data quality for SSEP analysis. Electrodes placed on the occipital and temporal regions were excluded from the analysis, as these areas are not primarily involved in somatosensory processing. The remaining signals were filtered using a Type II zero-phase Chebyshev filter with a passband of 1–30 Hz, following established protocols¹⁹. EEG signals were then segmented into epochs ranging from 500 ms before to 1000 ms after stimulus onset and re-referenced using the Common Average Reference (CAR) method. Trials affected by overlapping stimulations or other inconsistencies were discarded. To remove artifacts, Independent Component Analysis (ICA) was performed using the AMICA algorithm³⁰, which differs from conventional ICA by adopting an adaptive mixture model. This approach does not assume fixed spatial stationarity of EEG sources, making it particularly effective in capturing dynamic variations in both neural and non-neural components across different recording segments. The ICLabel classifier, integrated into EEGLAB, was employed to automatically categorize independent components into predefined artifact classes, such as eye movements and electrical interference. Automated classification was supplemented with manual inspection of scalp topographies, temporal characteristics, and frequency spectra to refine the rejection of subject-specific artifacts, particularly those related to electrical stimulation. Baseline correction was applied by subtracting the mean pre-stimulus activity from each epoch. To account for slow drifts that could introduce misalignment in the signals, a Bayesian non-parametric smoothing method was applied on an epoch-by-epoch basis to estimate and remove these trends. SSEP waveforms were obtained through conventional averaging: epochs were grouped according to experimental conditions and the mean response was calculated across trials. Averaging was performed at both the individual level, to assess subject-specific SSEP, and the group level, combining data from all participants to obtain a general response (i.e., grand average).

Statistical analysis

Statistical analyses were performed using RStudio software (RStudio Team, 2015, Version 1.2.5001). All variables were assessed for normality using the Shapiro-Wilk test. When normality assumptions were violated for at least one variable, non-parametric tests were employed. Current intensity perception ratings were analyzed using a repeated measure analysis of variance (ANOVA) to test for differences between conditions. Pairwise t-tests were conducted as post hoc comparisons. To compare the SSEP amplitudes between conditions, the non-parametric Friedman’s rank sum test was applied to the N60, while ANOVA was used for the N100. For post hoc analysis, pairwise Wilcoxon signed-rank tests were performed for the N60, and pairwise t-tests were applied for the N100. All reported p-values were adjusted for multiple comparisons using the Bonferroni correction. Correlational analyses were performed between SSEP amplitudes and current intensity perception ratings using repeated measure correlation (i.e., *rmcorr*³¹).

Data availability

The data that support the findings of this study are not openly available due to reasons of sensitivity and are available from the corresponding author upon reasonable request. Data are located in controlled access data storage at University of Padova.

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M.B. Conceptualization, Data Curation, Investigation, Methodology, Software, Visualization, Writing - Original Draft; S.T. Data Curation, Formal Analysis, Investigation, Methodology, Software, Writing - Original Draft; M.P. Data Curation, Formal Analysis, Investigation, Software, Visualization, Writing - Original Draft; E.T. Investigation, Methodology, Software; W.T. Data Curation, Formal Analysis, Investigation, Software; A.B. Investigation, Methodology; P.B. Resources, Project Administration, Supervision; G.S. Data Curation, Resources, Methodology, Project Administration, Supervision; E.M. Funding Acquisition, Resources, Project Administration, Supervision; A.D.F. Conceptualization, Funding Acquisition, Resources, Methodology, Project Administration, Supervision; All the authors: Writing - Review & Editing.

Declarations

Competing interests

The authors declare no competing interests.

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